

Context-Aware Explainable AI Curator for Virtual Museums using Retrieval- Augmented Generation

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Introduction

- Virtual museums preserve and showcase cultural heritage using AR, VR, and AI.
- AI-powered virtual guides enable natural language interaction with exhibits.
- Retrieval-Augmented Generation (RAG) improves response accuracy by retrieving information from verified museum knowledge sources.
- Existing AI systems often lack transparency, making it difficult for users to trust the generated responses.
- Explainable AI (XAI) enhances trust by providing source attribution and reasoning behind AI-generated answers.
- Proposed Solution: Develop a Context-Aware Explainable AI Curator that combines AR, RAG, and XAI to deliver personalized, accurate, and explainable museum experiences.
- Each response includes:
 - 📌 Source exhibit
 - 📄 Reference document
 - 📊 Confidence score
 - 💡 Explainable reasoning

IMPORTANCE OF THE RESEARCH TOPIC

Digital museums are becoming increasingly popular due to their accessibility, educational value, and ability to reach global audiences. Visitors expect AI-powered museum guides to provide accurate, context-aware, and trustworthy information about artifacts and historical events.

Integrating RAG with Explainable AI offers several benefits:

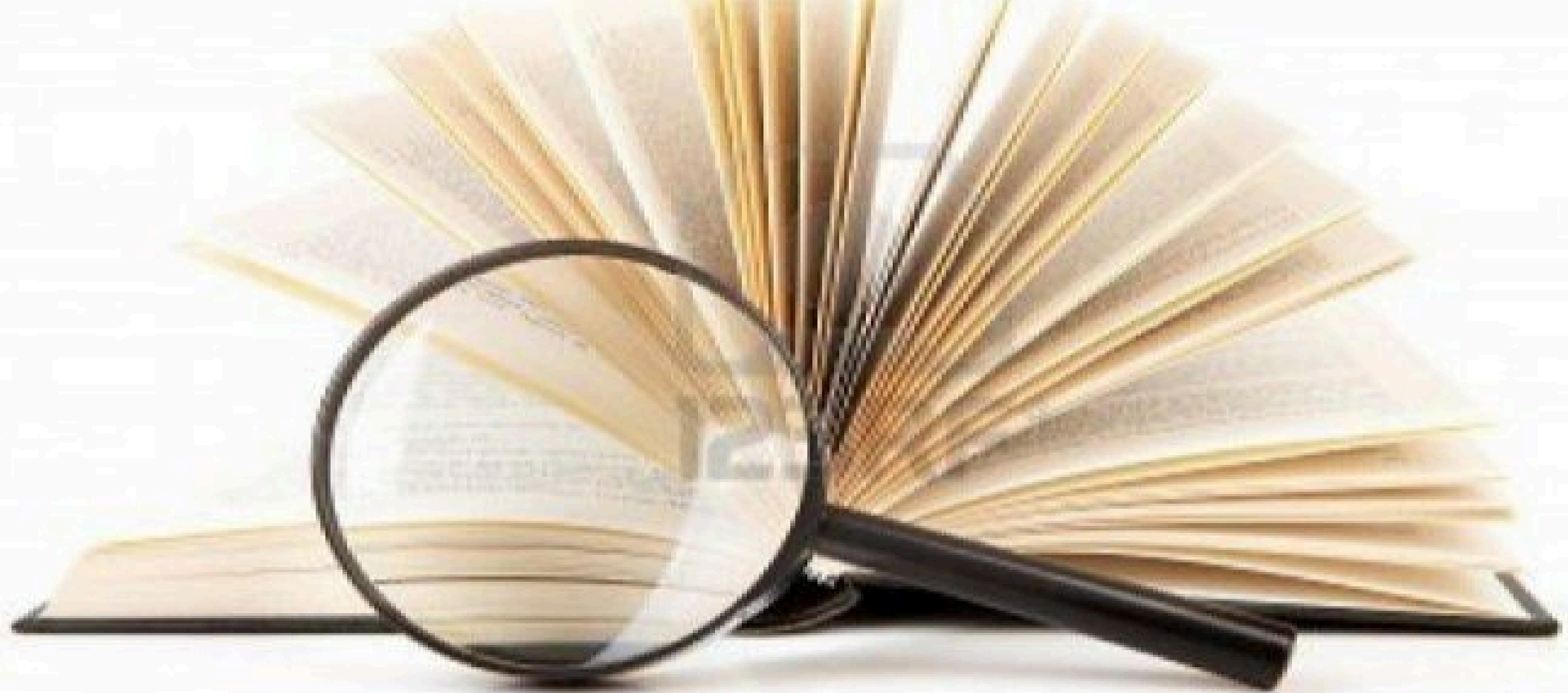
- Increases user trust by showing the evidence behind every response.**
- Reduces misinformation and AI hallucinations by grounding answers in verified museum documents.**
- Enhances learning by allowing visitors to trace information back to the original exhibit.**
- Supports museum curators by providing transparent AI assistance rather than opaque automated responses.**
- Improves visitor engagement through personalized, context-aware interactions while maintaining credibility.**

This research contributes toward building AI systems that are not only intelligent but also transparent, accountable, and suitable for cultural heritage applications.

CURRENT CHALLENGES IN THE SELECTED DOMAIN



- **Lack of Explainability:** Most conversational AI systems provide answers without indicating their source or reasoning process, making it difficult for users to verify information.
- **Hallucinated Responses:** Large Language Models may generate plausible but incorrect historical facts when reliable evidence is unavailable.
- **Limited Context Awareness:** Existing systems often fail to consider the visitor's current exhibit, interests, or conversation history, resulting in generic responses.
- **Insufficient Source Attribution:** Many museum chatbots do not reference the exhibit or document from which information was retrieved, reducing transparency.
- **Low User Trust:** Without confidence scores or explanations, users may hesitate to rely on AI-generated historical information.
- **Fragmented Integration:** While RAG and XAI have been individually explored, their combined implementation within immersive virtual museum environments remains limited.

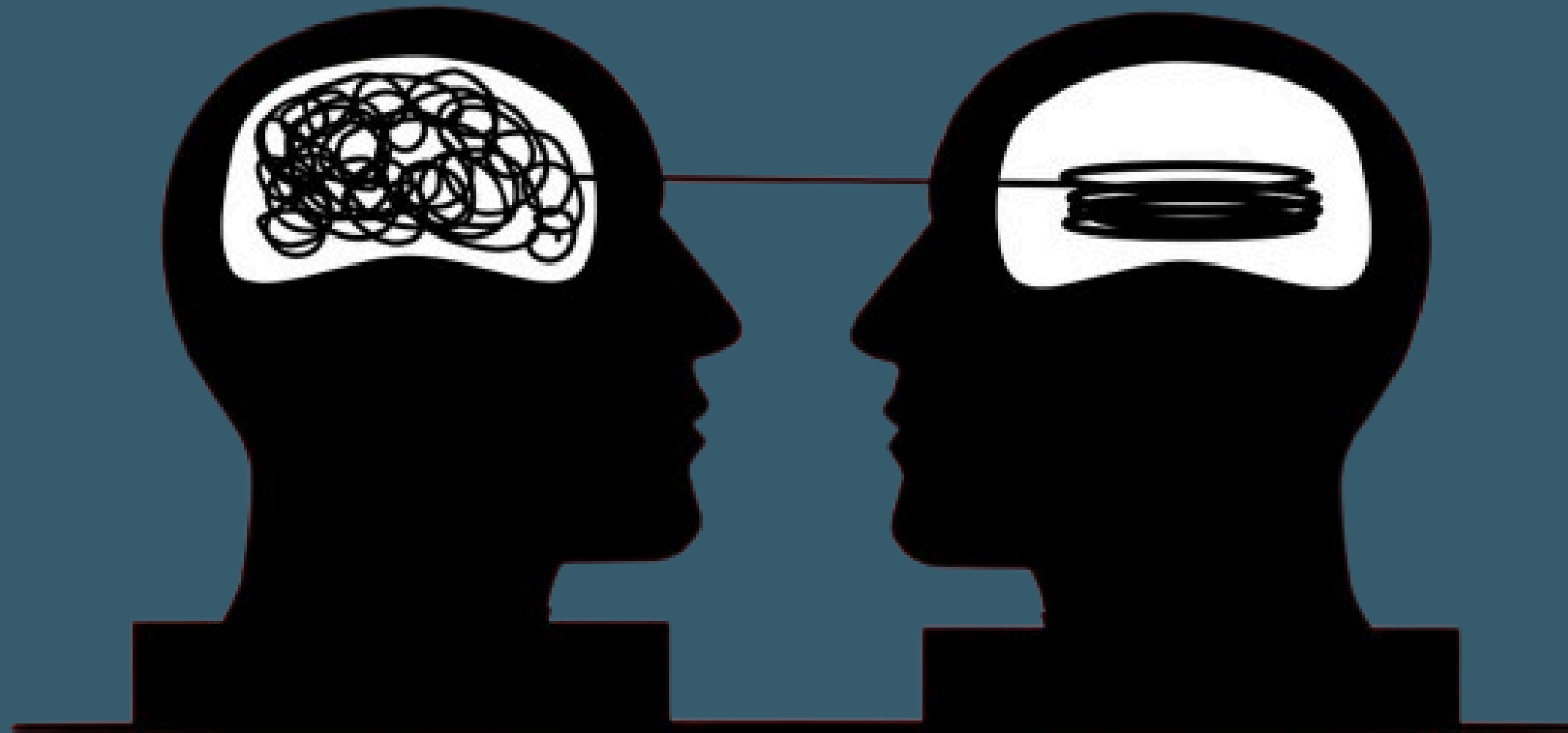


Literature Review

S. No.	Title & Citation	Methodology & Architecture	Dataset / Tools	Major Findings	Key Advantages	Identified Limitations
1	SA-RAG in Virtual Museums de Oliveira et al. (2025), ACM IMX '25	Combines eye-tracking (visual attention) with Retrieval-Augmented Generation (RAG) and LLM to provide context-aware VR museum guidance.	Exhibit metadata, 3D virtual museum, VR eye-tracking gaze data.	Gaze-guided retrieval improves contextual response accuracy over generic LLMs.	Personalized retrieval based on visitor attention; better contextual relevance.	Prototype-scale implementation; no confidence score, source citation, or explainability; requires eye-tracking hardware.
2	RAG Avatars for Virtual Archaeology Kerle-Malcharek et al. (2026), arXiv	Proposes a design framework (realm, personality, interaction) and develops a RAG-based VR archaeological avatar.	Curated archaeological texts, expert QA benchmark, VR workload questionnaires.	Avatar maintained topical conversations with below-average workload.	Reusable design framework with empirical evaluation.	Small evaluation dataset; lacks confidence scores and source attribution.
3	Image-to-3D Voice RAG Guidance Filho et al. (2026), IEEE AIxVR	Generates 3D artifacts from a single image and integrates voice-controlled RAG guidance in VR.	Exhibition photographs, voice commands.	Single-image reconstruction supports rapid virtual exhibit generation.	Multimodal interaction (voice + RAG); reduces content creation effort.	Lower 3D quality than multi-view reconstruction; no explainable responses or citations.
4	Textile Heritage AI Assistant Ariya et al. (2025), Computers & Education: XR	Mixed-methods evaluation of a multilingual voice-based AI docent in a VR textile museum.	Textile heritage dataset; 25 university students.	High user satisfaction and improved learning engagement.	Real user evaluation; multilingual support; preserves intangible heritage.	Small sample size; no traceability, explainability, or confidence estimation.
5	AI-Directed PCG in XR Museums Çakmak & Çekmiş (2026), Frontiers in VR	Rule-based AI Director adapts museum experiences based on visitor movement and gestures.	XR tracking data, Catmull–Rom splines.	Successfully classifies visitor behavior and adapts museum flow.	Strong behavioral personalization with museum constraints.	Does not use RAG or LLMs; lacks natural language interaction and explainability.
6	ARIA Adaptive AR Framework Konstantakis & Iakovaki (2026), Information (MDPI)	Conceptual framework combining multimodal context, affect recognition, and RAG for adaptive storytelling.	CIDOC-CRM heritage metadata standards.	Demonstrates how RAG and affect-awareness improve AR storytelling.	Ethical architecture with knowledge-grounded retrieval.	Conceptual only; no implementation, benchmark, or user study.
7	VR, AIoT & Semantics Museum Kostadimas et al. (2025), SMAP Workshop	Integrates VR, AIoT sensors, semantic web ontologies, and LLM-based guidance.	HMD behavioral traces, digital twin model.	Enables intelligent visitor profiling and semantic route optimization.	Combines semantic web with AI for structured retrieval.	Work-in-progress without quantitative evaluation or completed implementation.

S. No.	Title & Citation	Methodology & Architecture	Dataset / Tools	Major Findings	Key Advantages	Identified Limitations
8	Virtual Art & ML Curation Kumari et al. (2025), ShodhKosh	Systematic review of machine learning methods for virtual exhibition design.	Survey of ML-based curation literature.	ML improves artifact classification and thematic organization.	Connects ML techniques with digital exhibition planning.	Review paper only; no practical implementation or RAG integration.
9	3D Point-Cloud Restoration & VR Li (2024), Applied Mathematics and Nonlinear Sciences	Uses RBF interpolation, GC-GAN restoration, texture mapping, and VR visualization.	Dunhuang mural point clouds, SUS, UEQ.	Achieved high restoration accuracy (SSIM 0.83; PSNR 26.36 dB) with good usability.	Strong technical restoration and quantitative evaluation.	Focuses only on restoration; no conversational AI or explainability.
10	Museum in the Classroom (AR) Ly et al. (2025), CHI EA '25	AI chatbot integrated with curated artifact libraries in an educational AR application.	Curated artifact collections; iPad-based AR.	Makes museum experiences accessible in classrooms.	Educational accessibility without physical museum visits.	Classroom-focused; lacks VR, RAG source citation, and explainability.
11	RAG Survey for AIGC Zhao et al. (2024), arXiv	Comprehensive taxonomy of Retrieval-Augmented Generation architectures and evaluation methods.	Cross-domain RAG benchmarks and literature.	RAG effectively reduces hallucinations and outdated knowledge.	Strong technical foundation for retrieval-based AI systems.	Domain-general; not specific to museums or VR applications.
12	VR/AR in Museum Learning Meta-Analysis Zhou et al. (2022), Educational Research Review	PRISMA-based meta-analysis of 65 VR/AR museum learning studies.	65 published research articles.	VR/AR significantly improves learning outcomes and engagement.	Strong empirical evidence supporting immersive learning.	Conducted before LLMs and RAG; no AI explainability discussion.
13	The Invisible Museum Platform Zidianakis et al. (2021), Electronics (MDPI)	Authoring platform using CIDOC-CRM, EDM, and deep learning for semantic heritage management.	Digitized artifacts and curator narratives.	Enables standards-based creation of virtual museum exhibitions.	Strong semantic provenance using museum standards.	Focused on authoring, not conversational AI or explainable question answering.
14	AI & Museums: Collaboration Analysis Nguyen (2024), Journal of Community Archaeology	Qualitative analysis of AI adoption in museums using institutional case studies.	Museum case studies (British Museum, Louvre, etc.).	Identifies incomplete heritage data as a major challenge for AI accuracy.	Highlights practical deployment challenges and data quality issues.	No technical architecture or implemented AI system.

COMPARATIVE ANALYSIS



Ref.	Paper & Citation	Techniques / Algorithms	Architecture / Methodology	Dataset / Tools	Evaluation Metrics	Key Results	Advantages	Limitations
1	de Oliveira et al. (2025). <i>SA-RAG in Virtual Museums</i> . ACM IMX '25	Retrieval-Augmented Generation (RAG), Large Language Model (LLM), Eye-tracking, Embedding-based Retrieval	Visitor gaze is captured in VR using eye-tracking and incorporated into retrieval so the LLM generates responses grounded in exhibits	VR museum environment, exhibit metadata, eye-tracking gaze traces, LLM, vector retrieval	Contextual response accuracy; qualitative user evaluation	Gaze-aware retrieval produced more contextually relevant responses than a conventional LLM guide and improved personalization.	Personalizes retrieval using user attention; reduces irrelevant responses.	Prototype implementation; limited evaluation; no confidence score, source citation, or explainability.
2	Kerle-Malcharek et al. (2026). <i>RAG Avatars for Virtual Archaeology</i> . arXiv:2603.23353	RAG, Conversational Avatar, LLM, Expert Knowledge Retrieval	Defines avatar realm, personality, interaction modality, and retrieval pipeline. A RAG-powered avatar answers questions using curated	Maxentius Mausoleum corpus, scholarly documents, expert QA benchmark, VR workload questionnaire	User workload questionnaire, expert answer evaluation, topical relevance	Participants reported below-average workload while maintaining meaningful archaeological conversations.	Reusable architecture with empirical evaluation.	Small benchmark; limited domain coverage; lacks citations, confidence estimation, and explanations.
3	Filho et al. (2026). <i>Image-to-3D Voice RAG Guidance</i> . IEEE AIXVR	Single-image 3D Reconstruction, Voice Interface, RAG	Converts a single exhibit image into a 3D object and integrates a voice-controlled RAG assistant.	Exhibition photographs, speech input, 3D generation pipeline, RAG retrieval system	Qualitative evaluation	Acceptable virtual exhibits generated from single images with natural voice-based retrieval.	Simplifies exhibit creation; multimodal interaction (voice + vision + RAG).	Lower reconstruction quality than multi-view methods; lacks explainability.
4	Ariya et al. (2025). <i>Textile Heritage AI Assistant</i> . Computers & Education: X Reality	Multilingual Conversational AI, Voice Recognition, VR Museum Assistant	Mixed-method evaluation of an AI docent integrated into a VR reconstruction supporting multilingual voice interaction.	Lamphun brocade textile dataset, VR environment, 25 students	Likert-scale user satisfaction	Visual quality = 4.40/5; User experience = 4.20/5.	Real-world evaluation; multilingual interaction; preserves cultural heritage.	Small participant group; no explainable AI, confidence scores, or source attribution.
5	Çakmak & Çekmiş (2026). <i>AI-Directed Procedural Content Generation in XR Museums</i> . Frontiers in Virtual Reality	Rule-based AI Director, Visitor Behavior Classification, Procedural Content Generation (PCG)	Uses movement and gesture data to classify visitors and dynamically adapt exhibit sequencing.	XR tracking, controller movement, hand/head pose, Catmull–Rom splines	Qualitative behavioral classification	Successfully adapted museum flow according to visitor profiles.	Dynamic personalization while preserving museum semantics.	No conversational AI or RAG; rule-based; lacks quantitative evaluation.

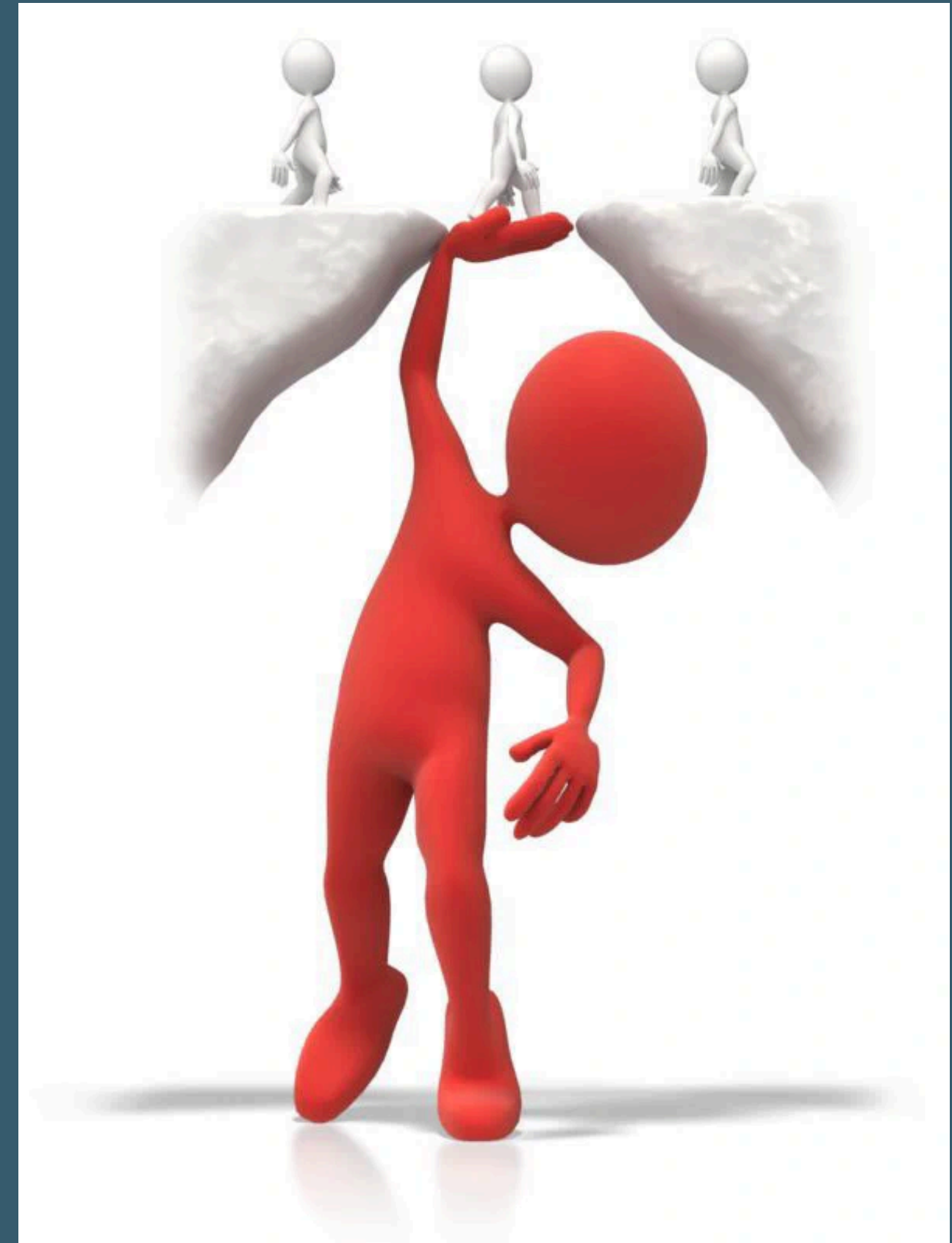
Ref.	Paper & Citation	Techniques / Algorithms	Architecture / Methodology	Dataset / Tools	Evaluation Metrics	Key Results	Advantages	Limitations
6	Konstantakis & Iakovaki (2026). <i>ARIA Adaptive AR Framework</i> . Information (MDPI)	RAG, Context Awareness, Affect Recognition, Semantic Knowledge Graphs	Conceptual framework integrating multimodal context, emotional recognition, heritage ontologies, and RAG.	CIDOC-CRM heritage metadata, semantic knowledge bases	NR (Conceptual)	Shows contextual information with grounded retrieval can improve adaptive storytelling.	Integrates ethics, personalization, and semantic standards.	No implementation or evaluation.
7	Kostadimas et al. (2025). <i>VR, AIoT & Semantics Museum</i> . SMAP Workshop	AIoT, Semantic Web Ontologies, Digital Twin, LLM Assistance	Integrates IoT sensors, semantic web technologies, VR, and LLMs for visitor profiling.	Digital twin, AIoT sensors, HMD traces, semantic ontologies	NR (Work-in-progress)	Proposed semantic visitor profiling and intelligent route optimization.	Strong semantic representation and interoperability.	Preliminary framework without implementation or validation.
8	Kumari et al. (2025). <i>Virtual Art & Machine Learning Curation</i> . ShodhKosh	Machine Learning Review, Image Classification, Style Transfer, Similarity Learning	Systematic review of ML techniques for artwork classification and exhibition design.	Published ML literature	NA (Review)	ML improves artwork categorization and thematic exhibition organization.	Comprehensive overview of ML applications.	No implementation; excludes RAG and Explainable AI.
9	Li (2024). <i>3D Point-Cloud Restoration & VR</i> . Applied Mathematics and Nonlinear Sciences	RBF Interpolation, GC-GAN Restoration, Texture Mapping	Repairs damaged point clouds using interpolation and GAN-based restoration before VR visualization.	Dunhuang mural point clouds, VR platform	SSIM, PSNR, SUS, UEQ	SSIM = 0.83; PSNR = 26.36 dB; SUS = 73.875.	Strong quantitative restoration evaluation.	Focuses on restoration; no conversational AI or explainability.
10	Ly et al. (2025). <i>Museum in the Classroom (AR)</i> . CHI EA '25	AI Chatbot, Augmented Reality, Educational Conversational Interface	Combines curated digital artifacts with AI chatbot in an iPad-based AR application.	Curated artifact library, AR classroom application	Qualitative classroom observations	Provides interactive museum learning without physical visits.	Accessible educational deployment.	AR-focused; lacks RAG and source citation.

Ref.	Paper & Citation	Techniques / Algorithms	Architecture / Methodology	Dataset / Tools	Evaluation Metrics	Key Results	Advantages	Limitations
11	Zhao et al. (2024). <i>Retrieval-Augmented Generation Survey for AIGC.</i> arXiv	RAG, Dense Retrieval, Sparse Retrieval, Hybrid Retrieval, LLMs	Surveys retriever-generator coupling, indexing strategies, retrieval methods, and evaluation benchmarks.	Cross-domain benchmark datasets	Comparative benchmark analysis	RAG significantly reduces hallucinations and improves factual accuracy over standalone LLMs.	Strong technical foundation for retrieval-based AI.	General survey; not museum-specific.
12	Zhou et al. (2022). <i>VR/AR in Museum Learning Meta-analysis.</i> Educational Research Review	PRISMA Meta-analysis	Synthesizes findings from museum learning studies evaluating VR/AR effectiveness.	65 peer-reviewed museum studies	Meta-analysis effect sizes	VR/AR significantly improves learning achievement, engagement, and motivation.	Strong empirical evidence for immersive learning.	Published before RAG and generative AI.
13	Zidianakis et al. (2021). <i>The Invisible Museum Platform.</i> Electronics (MDPI)	CIDOC-CRM, Europeana Data Model (EDM), Deep Learning Ontology Mapping	Authoring platform enabling standards-compliant virtual exhibitions using semantic technologies.	Digitized artifacts, curator narratives, CIDOC-CRM, EDM	Qualitative validation	Enables semantically rich virtual exhibitions with preserved artifact context.	Strong semantic provenance and interoperability.	Not a conversational AI system; predates RAG and Explainable AI.
14	Nguyen (2024). <i>AI & Museums: Collaboration Analysis.</i> Journal of Community Archaeology	Qualitative Analysis, Case Study Review	Reviews AI adoption in museums, including virtual guides and reconstruction.	Case studies (British Museum, Louvre, etc.)	Qualitative discussion	Identifies incomplete heritage datasets as a major challenge for reliable AI.	Highlights practical deployment challenges.	No technical implementation or experimental validation.

PROBLEM STATEMENT

- Existing virtual museums provide AI-based explanations but often operate as black-box systems.
- AI responses lack transparency, making it difficult for users to verify information.
- Limited context awareness, with minimal consideration of user history, exhibit location, and conversation context.
- Absence of source attribution, confidence scores, and explainable reasoning reduces user trust.
- Most existing solutions focus on either personalization or immersion, but not both with explainability.
- Few systems integrate AR, RAG, Context Awareness, and Explainable AI into a unified platform.
- Need: Develop a Context-Aware Explainable AI Curator that delivers accurate, personalized, and evidence-backed responses with source citations and confidence scores to enhance trust and learning.

RESEARCH GAP



Research Gap	Limitations in Existing Studies	How the Proposed Work Addresses the Gap
Lack of Explainability	Most AI-based museum guides generate responses without explaining the reasoning process or retrieved evidence.	Integrates Explainable AI (XAI) to provide clear "Why this answer?" explanations for every generated response.
Absence of Source Attribution	Existing systems rarely identify the exhibit, document, or knowledge source used to generate responses.	Displays the exact source exhibit, museum catalogue entry, and retrieved document supporting each answer.
No Confidence Estimation	Users cannot determine the reliability or certainty of AI-generated responses.	Computes and presents a confidence score based on retrieval relevance and response generation quality.
Limited Context Awareness	Most systems rely on only one contextual factor, such as gaze, movement, or voice.	Combines exhibit location, user interaction, conversation history, and retrieval context to generate personalized responses.
Limited Integration of RAG and XAI	Existing research investigates Retrieval-Augmented Generation (RAG) and Explainable AI (XAI) independently rather than as a unified framework.	Develops an integrated framework that combines context-aware RAG with Explainable AI to improve transparency and accuracy.
Prototype or Conceptual Systems	Several studies present conceptual frameworks or prototype implementations with limited validation.	Implements and experimentally evaluates a functional prototype in a virtual museum environment.
Limited Practical Evaluation	Many studies use small datasets, limited user studies, or qualitative evaluations.	Evaluates the proposed system using realistic museum datasets and user-based performance metrics such as accuracy, trust, explainability, and user satisfaction.
Trust and Transparency Issues	Users are expected to trust AI-generated responses without verification or supporting evidence.	Enhances transparency through evidence-backed responses, confidence scores, source attribution, and explainable reasoning.

SUMMARY OF THE RESEARCH GAP

The existing literature demonstrates that Retrieval-Augmented Generation (RAG) improves the factual accuracy of AI-generated responses, while Virtual Reality (VR) enhances visitor engagement and context-aware techniques enable personalized museum experiences. However, most current systems focus primarily on generating responses and provide limited transparency regarding how those responses are produced. Key limitations include the lack of Explainable AI (XAI), absence of source attribution, unavailability of confidence estimation, and limited integration of multiple contextual signals into the retrieval process.

Additionally, many existing solutions remain conceptual or prototype-based and have been evaluated using small datasets or limited user studies, restricting their practical applicability and reducing user trust.

The proposed Context-Aware Explainable AI Curator for Virtual Museums using Retrieval-Augmented Generation addresses these research gaps by integrating context-aware retrieval, Explainable AI, source attribution, and confidence scoring into a unified end-to-end framework. By enabling users to understand not only what the AI answers but also why the answer was generated and which knowledge source supports it, the proposed system aims to improve transparency, reliability, user trust, and the overall educational experience in virtual museum environments.

Research Objectives



- Study existing virtual museum systems.
- Design an AI-powered virtual museum.
- Integrate AR with Retrieval-Augmented Generation.
- Develop a context-aware recommendation system.
- Improve response accuracy using trusted sources.
- Provide explainable AI-based responses.
- Evaluate system performance.

Proposed Methodology



Input

- User Query
- Selected Artifact
- Museum Knowledge Base

Output

- AI Explanation
- Source-backed Response
- Personalized Recommendations

Overall Architecture

- AR-based mobile application
- FastAPI backend
- RAG-enabled AI Curator
- MongoDB & ChromaDB storage

Proposed Methodology

Dataset Collection / Selection

- Museum metadata
- Historical documents
- Research articles
- 3D artifact models

Data Preprocessing

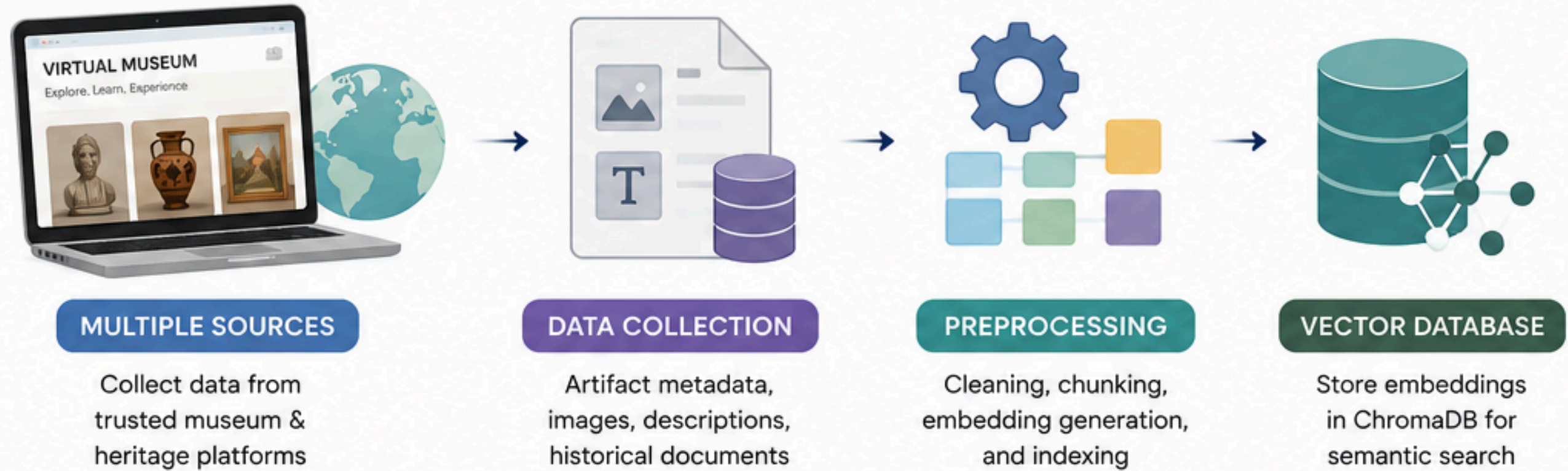
- Data cleaning
- Text chunking
- Embedding generation
- Vector database indexing

Algorithms / Models Used^{...}

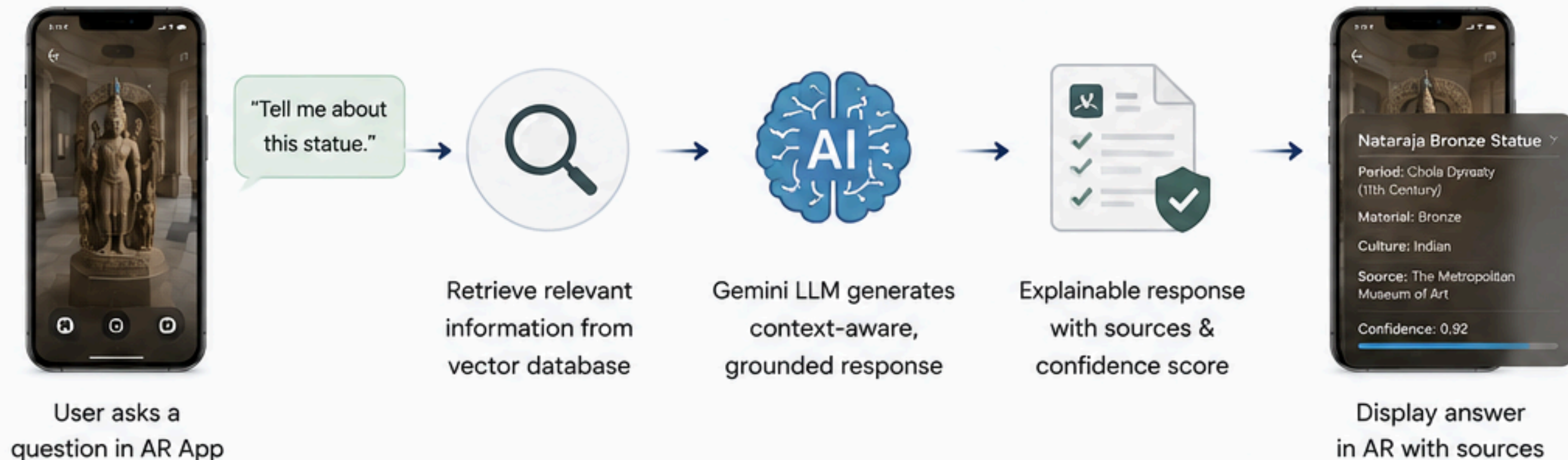
- AR Foundation
- Gemini API (LLM)
- (RAG)
- Sentence Transformers
- ChromaDB Vector Search
- Build knowledge base
- Generate embeddings
- Retrieve relevant documents

DATASET FOR AI CURATOR

Building the Knowledge Base for a Context-Aware Virtual Museum



HOW THE AI CURATOR USES THE DATA



DATASET SOURCES



Smithsonian Open Access

- Artifact metadata
- Descriptions
- Images



The Metropolitan Museum of Art Open Access

- 470,000+ artworks
- Metadata & images
- Historical information



Europeana API

- European heritage objects
- Cultural collections



Wikipedia / Wikidata

- Historical context
- Additional references



Sketchfab (CC Licensed)

- 3D models of artifacts
- For AR visualization



Knowledge Base

Unified, clean, and enriched dataset for accurate and meaningful AI interaction

Proposed Methodology

Software Requirements

- Unity 6
- FastAPI
- Python
- MongoDB Atlas
- ChromaDB
- VS Code

Hardware Requirements

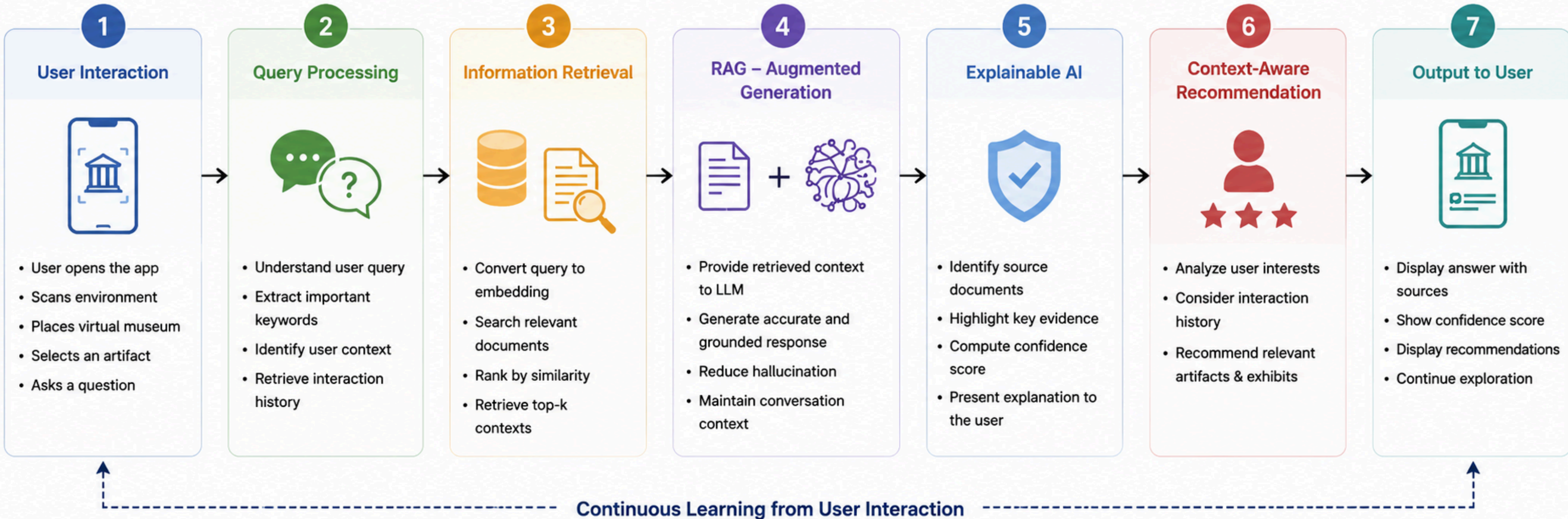
- ARCore-supported smartphone
- Laptop (8 GB+ RAM)

Evaluation Metrics

- Response Accuracy
- Response Time
- Retrieval Precision
- User Satisfaction

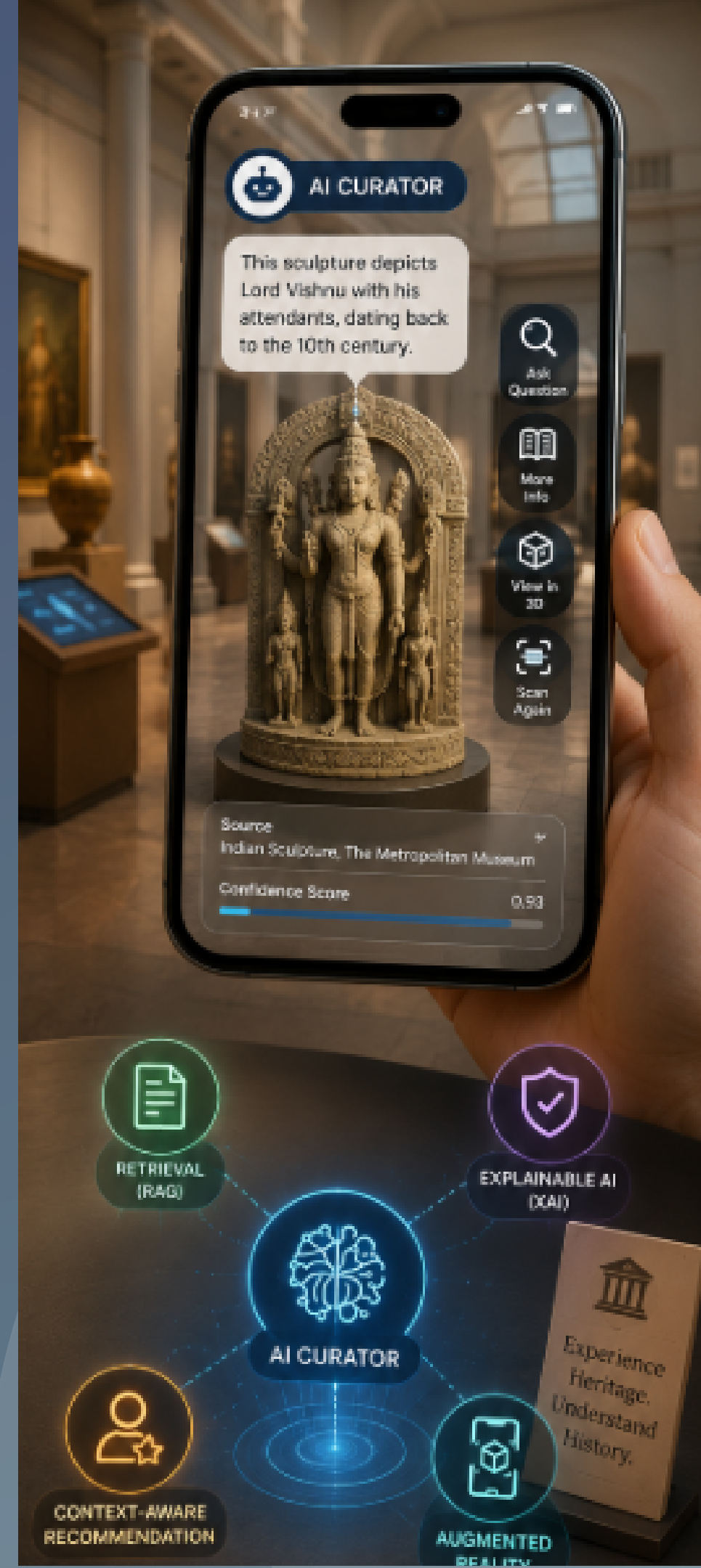


PROPOSED METHODOLOGY FLOW



Expected Outcomes

- Interactive smartphone-based AR museum
- Accurate AI-generated explanations
- Personalized artifact recommendations
- Improved user engagement
- Explainable and trustworthy AI responses
- Better accessibility to museum education



Work Completed



-  3D Museum: Immersive museum with Space, History & Natural History galleries.
-  AI Curator: Grounded RAG with confidence-based, explainable responses.
-  Voice Output: Real-time AI curator narration with Play/Stop controls.
-  Mobile AR: Real-world 3D artifact placement with touch interactions.
-  Cloud Deployment: Scalable frontend on Vercel and RAG backend on Railway.

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VIRTUAL HERITAGE MUSEUM

● Main Entrance Doorway

Enter AR Mode

First-Person Walk

Museum Map



Use WASD or Drag Mouse to Walk • Click any exhibit pedestal to inspect



VIRTUAL HERITAGE MUSEUM

Gallery 3: Smithsonian Paleontology Hall



Enter AR Mode

ART013

Gallery 3: Smithsonian Paleontology Hall



Triceratops horridus ("Hatcher")

Artifact Details

Ask AI Curator



CATEGORY
Ceratopsian Dinosaur
Paleontology



PERIOD
Late Cretaceous (c. 66
Million Years Ago)



Institution
National Museum of Natural History, Smithsonian
Institution



Origin
Lance Formation, Niobrara County, Wyoming, United
States



License
CC0 1.0 Universal (Smithsonian Open Access)

View on Official Smithsonian 3D



OFFICIAL CURATOR DESCRIPTION

The world's first mounted Triceratops skeleton, affectionately nicknamed 'Hatcher' (catalog USNM 4720). One of the most famous dinosaur fossils on Earth, boasting three massive brow and nasal defensive horns and an expansive solid bony neck frill.

Explore Artifact 360°

View Exhibit in Mobile AR

Ask AI Curator Questions

GALLERY 3

Smithsonian Paleontology Hall



TYRANNOSAURUS REX SKULL



TRICERATOPS HORRIDUS
("HATCHER")

ART013 | Late Cretaceous (c. 66 Million Years Ago)

N

Triceratops horridus ("Hatcher")

 Artifact Details

 Ask AI Curator

 AI MUSEUM CURATOR



CURRENTLY VIEWING:

Triceratops horridus ("Hatcher")

Gallery 3: Smithsonian Paleontology Hall

ART013

CURATOR

TONE:

 Educational

 Concise

 Friendly

 QUICK QUESTIONS:

What is this?

Tell me more

When was it created?

What was it used for?

Why is it important?

 AI Curator Response

It was mounted in 1905 for public display in the Smithsonian's Paleontology Hall, serving as the first Triceratops skeleton shown to visitors and as the definitive type reference for the ceratopsian clade.

 Listen

SOURCE

Triceratops horridus ("Hatcher") — Gallery 3: Smithsonian Paleontology Hall

 EVIDENCE FROM CURATED ARTIFACT RECORD

| *"Triceratops horridus ("Hatcher")*. Gallery: Gallery 3:

Ask about Triceratops horridus ("Hatcher")...

Ask 

Future Work

-  3D Hotspots: Interactive artifact annotations for detailed component information. ●●●
-  Guided Tours: Automated thematic tours with smooth camera navigation.
-  Artifact Comparison: Interactive side-by-side comparison of multiple artifacts.
-  Timeline: Chronological exploration connecting artifacts with historical events.
-  Personal Collection: Favorites and personalized museum recommendations.
-  Gamification: Achievement badges and rewards for visitor engagement.

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THANK YOU!

Questions & Suggestions
are Welcome



Heritage



AI



Immersion



Knowledge

“ Preserving the past, enriching the present,
inspiring the future. ”

